

Kari Lakshmi Apoorva

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SKILLS

- **Laboratory Skills:** Polymerase Chain Reaction (PCR), ELISA, DNA Extraction, Gel Electrophoresis, Blotting techniques
- **Tools/Platforms:** MS Powerpoint, MS Word, MS Excel
- **NCBI Resources:** PubMed, GenBank, Basic Local Alignment Search Tool (BLAST), Entrez
- **Soft Skills:** Strategic Mindset, Problem-Solving Skills, Team Player, Project Management, Adaptability

INTERNSHIP

- **Bihar State Milk Co-operative Federation** June 2023 – July 2023
Quality Control (QC) Summer Intern
 - Gained hands-on experience in physicochemical and microbiological analysis of raw milk to assess its quality and safety.
 - Assisted the Quality control department, laboratory technicians with standardization of laboratory chemicals and calibration of analytical reagents and solutions, ensuring compliance with FSSAI norms.
 - Gained exposure to cold chain management, sample preservation protocols, and standard operating procedures (SOPs) used in milk collection and processing units.
 - Learnt the handling of various equipments, instruments and their demonstrations, their usage to test quality of milk in various dairy industries.

PROJECTS

- **Capstone Project** August 2024 – May 2025
School of Bioengineering and Biosciences, Lovely Professional University
Thesis topic: "Synthesis of fish scale-based nanoparticles for photocatalytic applications."
Supervisor: Dr. Vineet Kumar
 - Developed a sustainable approach to synthesize nanoparticles from fish scales, leveraging their natural hydroxyapatite and Collagen content.
 - Characterized the synthesized nanoparticles using advanced analytical techniques, including SEM, TEM, FTIR, XRD, and zeta potential.
 - Investigated the photocatalytic potential of the fish scale-based nanoparticles via degradation kinetics of methylene blue dye for environmental, biomedical applications.
 - Evaluated their antimicrobial potential by conducting antibacterial assays using MIC techniques in Mueller Hinton broth against E.coli and S.aureus to explore their wound healing properties, pharmaceutical and biomedical applications.
- **Community Development Project** July 2022
Community Development Project
NGO: Padala Charitable Trust (PCT)
 - Created an online survey that aims to assess the awareness levels of students regarding adolescence issues and their careers.
 - Participated in various education related volunteer activities.
 - Designed PPT presentations in two different languages related to menstrual education
 - Participated in the ground level event of Project Varadhi, meant for the upliftment of orphans and semi Orphans of rural backgrounds.

CERTIFICATES

- Next Generation Sequencing (NGS) for Influenza and SARS-COV-2: From Sample Collection to Analysis September 2024
An online non-credit course offered by World Health Organization (WHO)
- Municipal Solid Waste Management in Developing Countries April 2024
An online non-credit course by Ecole Polytechnique Federale de Lausanne and offered through Coursera
- Precision Medicine March 2023
An online non-credit course authorized by University of Geneva, Coursera
- Introduction to Intellectual Property October 2022
An online non-credit course authorized by University of Pennsylvania, Coursera
- Introduction to Biology of Cancer May 2022
An online non-credit course authorized by Johns Hopkins University, Coursera

Extracurricular Activities

- **Project Expo-2025 | LPU** April 2025
 - Presented a scientific poster based on my research work focused on the synthesis of fish scale based nanoparticles and their characterization.
- **International Conference on Biotechnology and Biomedical research (ICBBR-2024)** September 2024
 - Gained insights into innovative trends in environmental, industrial, and medical biotechnology.
 - Engaged with current research and emerging through expert lectures and technical lessons.

Celesta: A 24 hour Hackathon

March 2024

- Participated in a 24 hour hackathon organized by school of Bioengineering and Biosciences where innovative solutions for waste segregation and disposal have been provided along with a presentation

Workshop on Molecular Biology and Biochemistry Techniques | IIT Delhi

February 2024

- Participated in workshop based on Sequencing Techniques, Docking Methods and about Bioinformatic Tools
- Learnt about MEGA software, Autodock and Phylogenetic tree construction

AIU's second International Youth Festival – 'One World' | LPU

April 2023

- Participated in the event "One World" with the theme – Travel, Culture, and Tourism representing Afghanistan

Freshmen Talent Mining – Debate Competition (War Of Words) | LPU

September 2021

- Secured second place in the debate competition on the topic of animal testing, shared balanced perspectives on ethics, science, and alternatives in biotechnology

EDUCATION

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- **Lovely Professional University** Punjab, India
Bachelor of Technology – Biotechnology; CGPA: 7.8 August 2021 - Present
 - **Sri Tirumala Junior College** Andhra Pradesh, India
Intermediate; Percentage: 95.6% June 2019 - March 2021
 - **Vidyanjali School** Andhra Pradesh, India
Matriculation; Percentage: 90.6% June 2018 - June 2019